

20261006 Halpha Dispersion along Infall Stages

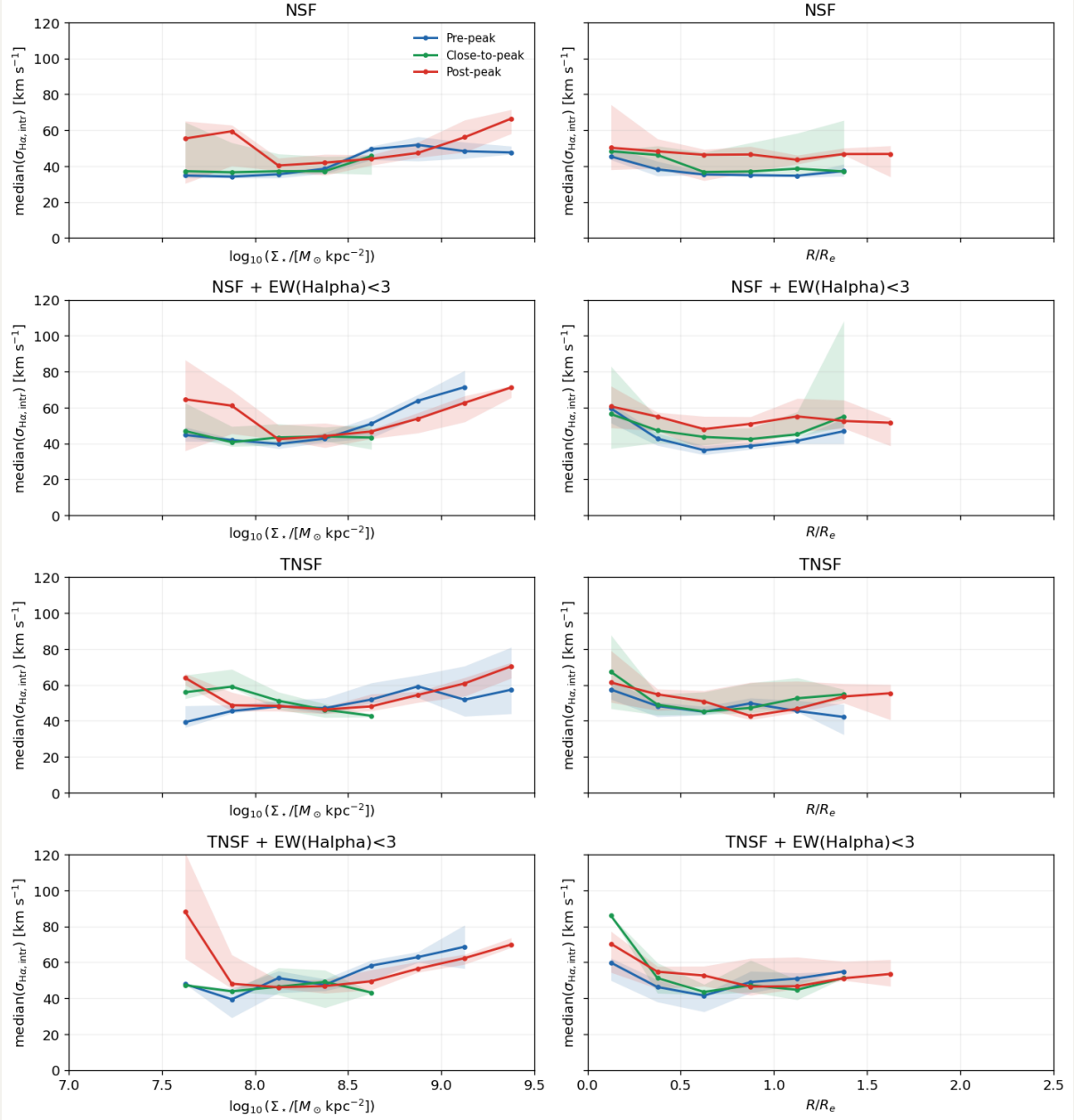
First, i check that even after the `SNR_POSTFIT`>25 cut, we still have some NSF post-peak spaxels that extend lower than stellar mass less than 7.5 or radius even outside $2 R_e$, and even goes up to 500km/s in velocity dispersion. In that unreliable regime, those are driven by only a few bins but contains many spaxels. So i further implement the following cuts to exclude them: in either mass or radial bin, it requires at least 25 distinct spatial bins and at least from galaxies.

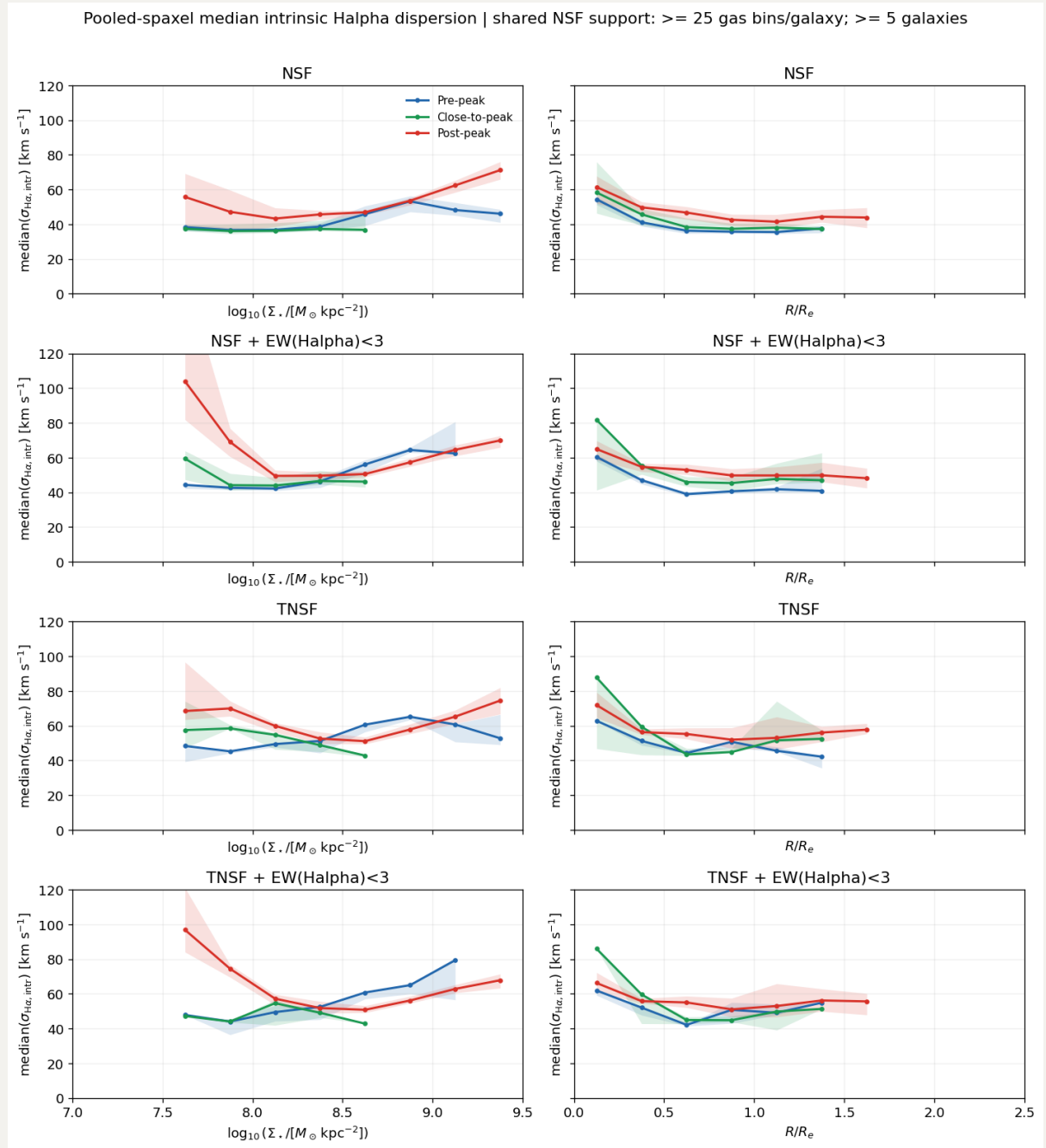
Then here we further make a selection on NSF. So our definition of NSF is $\neg((\text{HII}) \cap (\text{EW}(\text{H}\alpha) > 6\text{\AA}) \cap (\sigma(\text{H}\alpha) < 45\text{km/s}))$, then we can further select the "true" NSF ("truely" not power by young star photonization) by excluding the `COMP` region in NII BPT.

And in either NSF or TNSF regions, we can still add an additional selection of $\text{EW}(\text{H}\alpha) < 3\text{\AA}$ to select the HOLMES, the ionization by old stars.

As usual, we have equal-galaxy and pool-galaxy plots. And for equal-galaxy weighted median, i dont think there exists any trend across infall stages here; while for pooled-spaxel case i will say it is hard to tell.

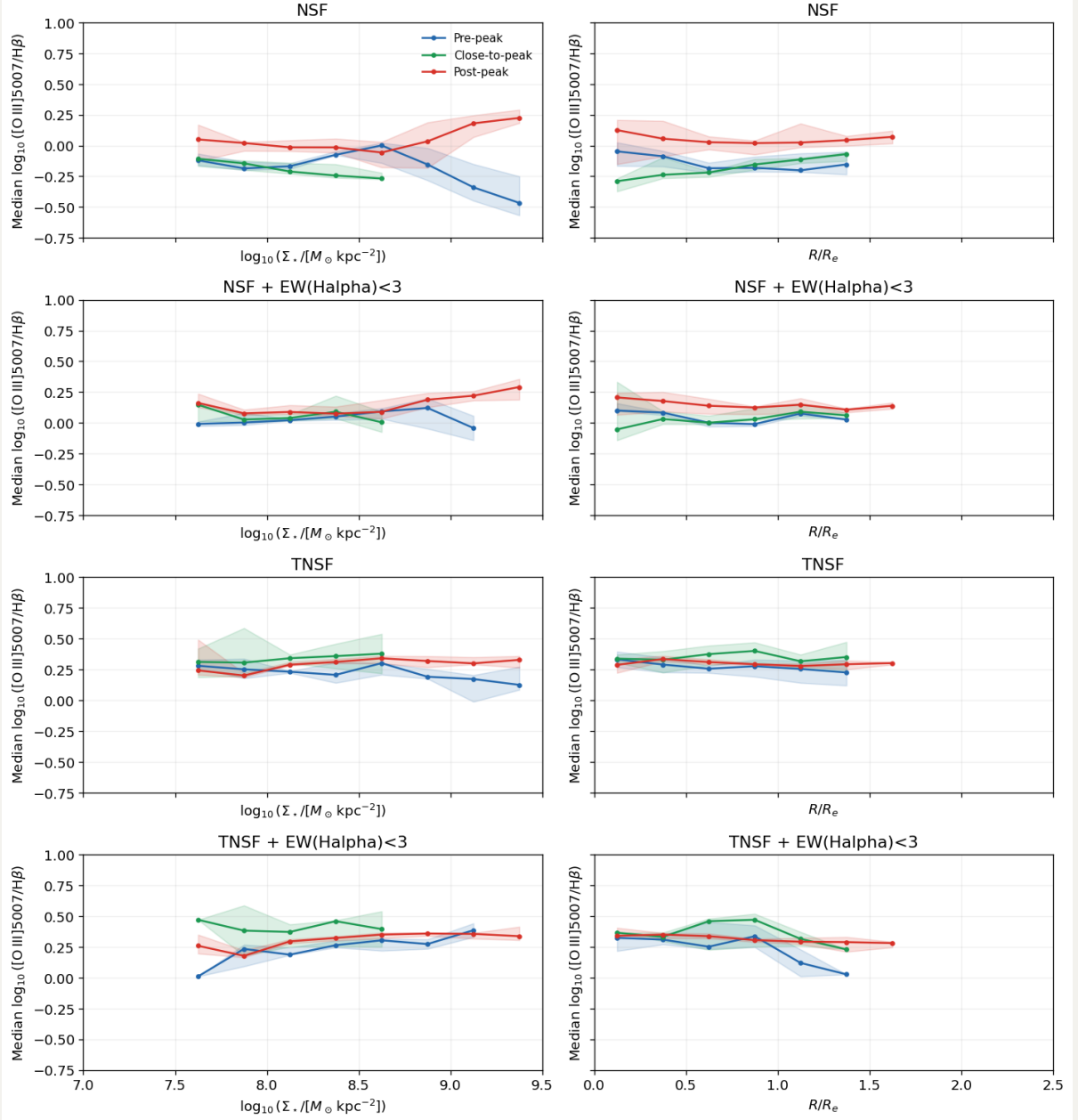
Equal-galaxy median intrinsic H α dispersion | shared NSF support: ≥ 25 gas bins/galaxy; ≥ 5 galaxies



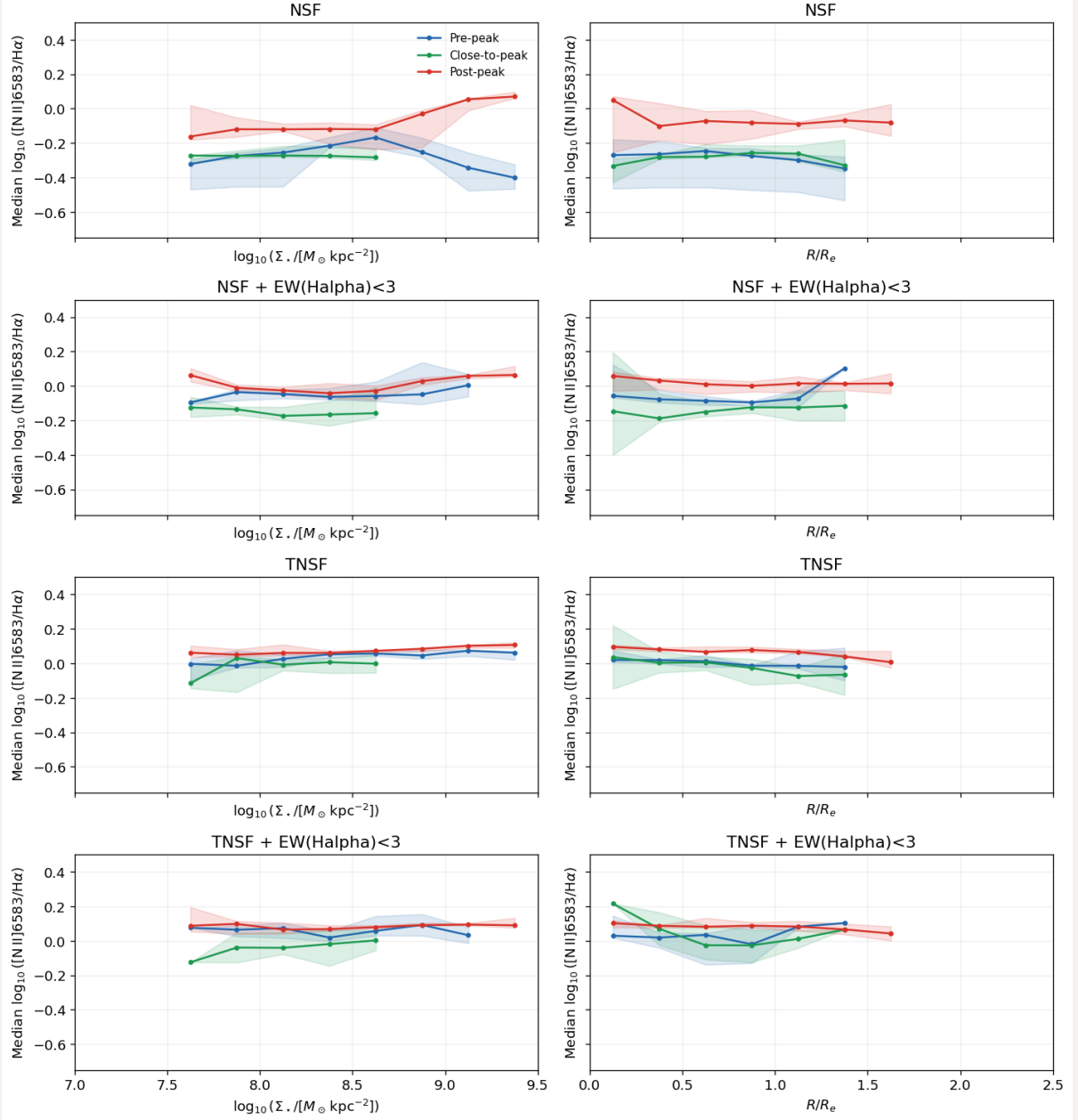


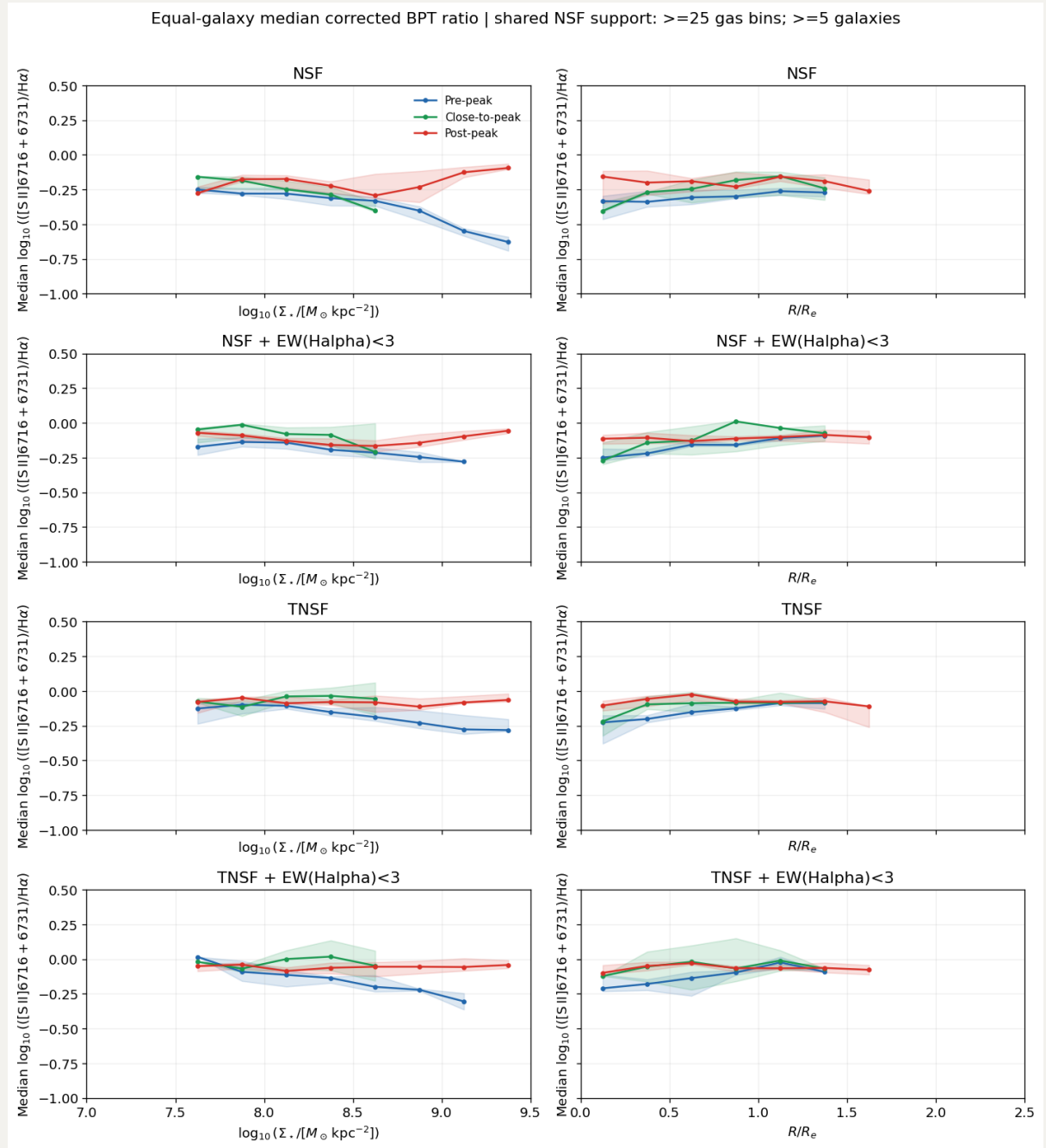
Also, we move back to BPT line ratios, it still shows that post-peak galaxies tend to have higher line ratios than pre-peak. And with stricter selection or even further EWH α cut, it seems to have less differences.

Equal-galaxy median corrected BPT ratio | shared NSF support: ≥ 25 gas bins; ≥ 5 galaxies



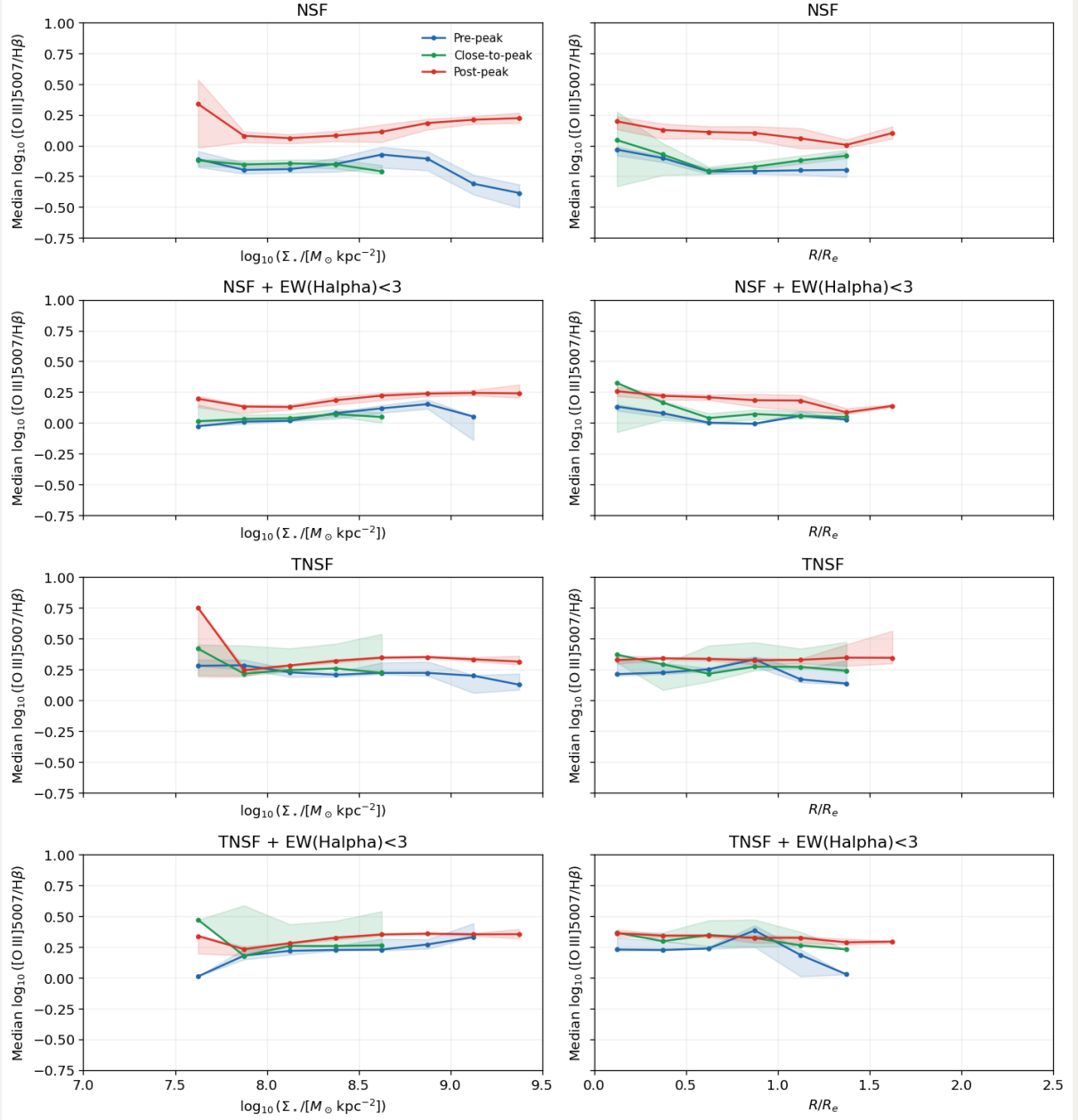
Equal-galaxy median corrected BPT ratio | shared NSF support: ≥ 25 gas bins; ≥ 5 galaxies



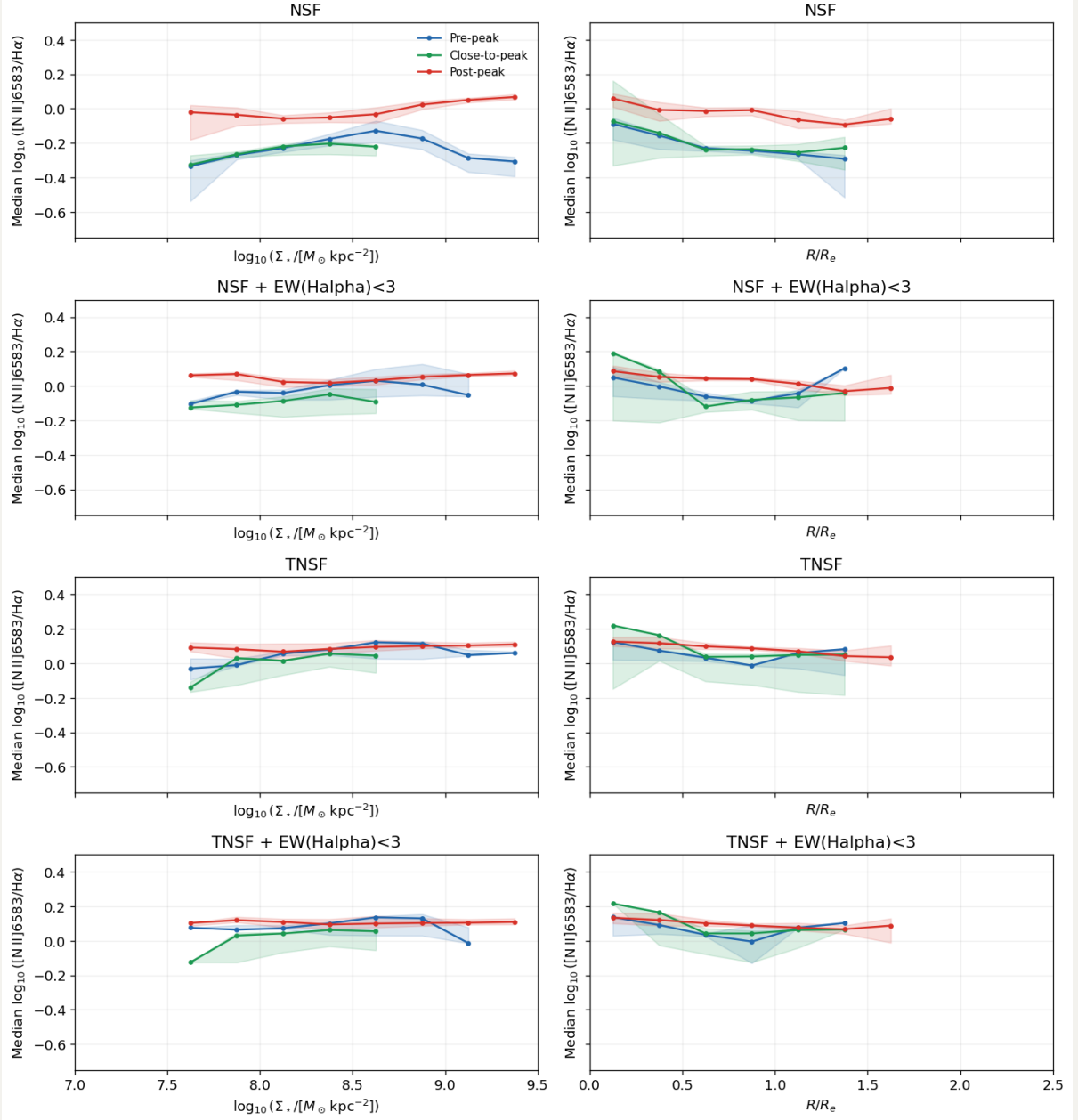


Similarly, we have pooled-spaxels case and it shows the same results.

Pooled-spaxel median corrected BPT ratio | shared NSF support: ≥ 25 gas bins; ≥ 5 galaxies



Pooled-spaxel median corrected BPT ratio | shared NSF support: ≥ 25 gas bins; ≥ 5 galaxies



Pooled-spaxel median corrected BPT ratio | shared NSF support: ≥ 25 gas bins; ≥ 5 galaxies

